

Session 01: Installation and Java Intro

Differentiation of coding languages:

1. Object Structured Programming Language: C, python
2. Object Based Programming Language: VB, VBScript, python
3. Object Oriented Programming Language: c++, Java, C#, python

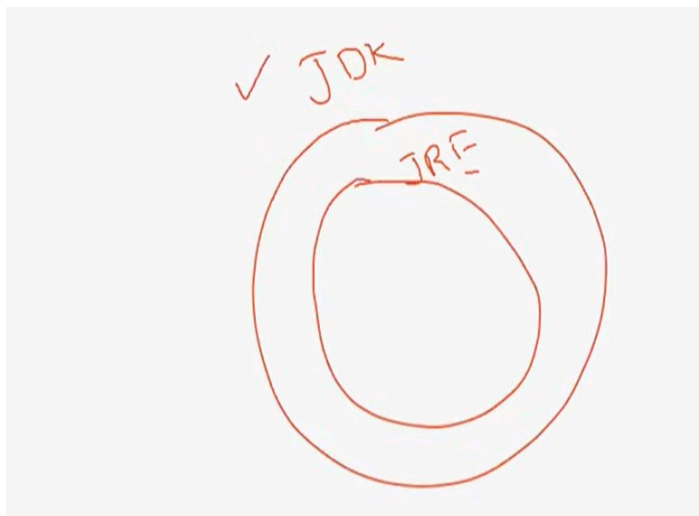
Note: Java is a statistically typed programming language

Features of Java:

1. Platform dependent
2. Case sensitive

Components:

1. JDK - Java Development kit
2. JRE- Java Run Time Environment
3. JVM - Java Virtual Machine



Session 02: Variables and Data Types

Variable: Variable is a container used to store the data

Ex: int Age= 25

Int x= 100

float itemprice=20.75

String name= "Ramu";

Char Grade ="A"

Data Type: Meaning what is the type of data stored in a Variable

Ex: String age =25

Different data Types:

1. **Primitive data type** : can hold only one value at one time (But with indexing we can save. Ex: `int a[] = new int[100]`)
2. **Non - Primitive data type/derived/Collections**: can store multiple values/characters

1. Primitive data types: Lowercase

byte, short, int, long -> Number without decimals

float , double -> decimal numbers

Char -> single character

Boolean -> true/false

Byte: to store 1 byte (from -128 to 127)

Short: to store 2 bytes (from -32,768 to 32,767)

int: 4 bytes

long 8 bytes

float: stores 4 bytes (functional numbers upto 7 decimal positions)

double: 8 bytes (functional numbers upto 15 decimal positions)

Char -> 2 byte single character

Boolean -> 1 byte true/false

2. Non - Primitive data type/derived/Collections: uppercase

String

ArrayList

HashMap

HashSet